Operating Systems (CSC-330) Instructor: Noreen Khalid

Assignment – 2 Fall 2025
Date Due: 9th Nov, 2025

Date assigned: 1st Nov, 2025
Marks: 10

Assignment Instructions:

- Work **individually**: plagiarism will result in zero marks.
- Solve this assignment on **Notebook**, take pics and upload on **GCR** before deadline.
- Attempt all questions.
- After Deadline **NO** assignment will be acceptable.

Topics: (Process Scheduling).

[Mapped with CLO-02 & 03]

Learning Objectives

By completing this assignment, You will be able to:

- Understand the working principles of CPU scheduling algorithms.
- Calculate **waiting time**, **turnaround time**, and **average times**.
- Differentiate between **preemptive** and **non-preemptive** scheduling.
- Simulate process scheduling scenarios manually or using a program.

Question 1: FCFS (First-Come, First-Served) – Non-Preemptive

Process	Arrival Time	Burst Time
P1	0	5
P2	2	3
P3	4	1

Tasks:

1. Draw the **Gantt Chart**.
2. Calculate **Waiting Time** and **Turnaround Time** for each process.
3. Compute the **Average Waiting Time** and **Average Turnaround Time**.

Question 2: SJF (Shortest Job First) – Non-Preemptive

Process	Arrival Time	Burst Time
P1	0	7
P2	2	4
P3	4	1
P4	5	4

Tasks:

1. Determine the **execution order** using non-preemptive SJF.
2. Draw the **Gantt Chart**.
3. Find **Waiting Time** and **Turnaround Time** for each process.

Hint: Select the shortest available process after the current one finishes.

Question 3: SJF (Shortest Remaining Time First) – Preemptive

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Tasks:

1. Draw the **Gantt Chart** showing all preemptions.
2. Calculate each process's **completion time**, **waiting time**, and **turnaround time**.
3. Compare average waiting times with non-preemptive SJF.

Hint: Always choose the process with the **shortest remaining time** among available ones.

Question 4: Round Robin Scheduling

Process	Arrival Time	Burst Time
P1	0	5
P2	1	4
P3	2	2
P4	3	1

Time Quantum (Q) = 2 units

Tasks:

1. Draw the **Gantt Chart** for Round Robin scheduling.
2. Compute **Waiting Time** and **Turnaround Time** for each process.
3. Find **Average Waiting Time** and **Average Turnaround Time**.

Hint: Each process gets a CPU for 2 units per turn, then the next process in the queue executes.

Question 5: Comparison & Reflection

Write short answers:

1. Which algorithm gives the **lowest average waiting time** in your examples? Why?
2. Which algorithm is **best for real-time systems** and why?
3. Define **starvation** and mention which algorithm may suffer from it.
4. Discuss how **time quantum** affects Round Robin performance.

Formulas Reference (Q-01 – Q-05)

- **Turnaround Time (TAT)** = Completion Time – Arrival Time
- **Waiting Time (WT)** = Turnaround Time – Burst Time
- **Average Waiting Time** = Σ WT / n
- **Average Turnaround Time** = Σ TAT / n

Question 6: Mixed Arrival with FCFS & Context Switching

Process	Arrival Time	Burst Time
P1	0	10
P2	3	5
P3	5	8
P4	9	6

Context Switching Time: 1 unit

Tasks:

1. Draw the **Gantt Chart** including context switching intervals.
2. Calculate:
   - Completion Time
   - Turnaround Time
   - Waiting Time
3. Compute **CPU Utilization** considering context switch overhead.
4. Compare results **with and without context switching**.

Hint: Add 1 time unit every time the CPU switches to another process.

Question 7: SJF (Preemptive) with Same Arrival Times

Process	Arrival Time	Burst Time
P1	0	6
P2	0	8
P3	0	7
P4	0	3

Tasks:

1. Draw the **Gantt Chart**.
2. Compute **Average Waiting** and **Turnaround Times**.
3. Compare with **SJF Non-Preemptive** using the same data.
4. Discuss the effect of preemption on average waiting time.

Hint: When all arrival times are equal, SJF preemptive behaves differently only if a shorter job arrives later during execution.

Question 8: Round Robin with Unequal Burst Times

Process	Arrival Time	Burst Time
P1	0	9
P2	1	5
P3	2	3
P4	3	7
P5	4	4

Quantum (Q): 3 units

Tasks:

1. Construct the **Gantt Chart** step by step (show ready queue state after each cycle).
2. Calculate **Waiting Time** and **Turnaround Time** for each process.
3. Analyze how changing Q to **2** or **5** would affect average waiting time.
4. Discuss the trade-off between **response time** and **context switching overhead**.

Question 9: Multi-Level Queue Scheduling

System uses **two ready queues**:

- **Queue 1:** Priority processes (High Priority, Quantum = 3, Round Robin)
- **Queue 2:** Normal processes (Low Priority, FCFS)
- **Scheduling Rule:** Queue 1 always executes before Queue 2.

Process	Queue	Arrival Time	Burst Time
P1	Q1	0	5
P2	Q2	1	7
P3	Q1	2	3
P4	Q2	3	4
P5	Q1	4	6

Tasks:

1. Create a **combined Gantt Chart** showing both queues' executions.
2. Compute **Average Waiting** and **Turnaround Time** for each queue separately.
3. Determine **overall CPU utilization** and **throughput**.
4. Discuss fairness: Does this policy cause starvation? How could you fix it?

Question 10: Priority Scheduling (Preemptive)

Process	Arrival Time	Burst Time	Priority
P1	0	4	2
P2	1	3	1
P3	2	5	4
P4	3	2	3
P5	4	1	5

(Lower number = higher priority)

Tasks:

1. Draw the **Gantt Chart**.
2. Calculate **Average Waiting Time** and **Turnaround Time**.
3. Discuss how the CPU changes when a higher-priority process arrives.
4. Compare results with **Non-Preemptive Priority Scheduling**.

Key Formulas Reference (Q. 06 – Q.10)

Metric	**Formula**
Turnaround Time (TAT)	Completion Time – Arrival Time
Waiting Time (WT)	TAT – Burst Time
Response Time (RT)	Time from Arrival → First Execution
CPU Utilization	(Busy Time / Total Time) × 100
Throughput	No. of Processes Completed / Total Time

Reflection Questions

1. Which scheduling algorithm provides the **best balance** between waiting time and response time?
2. How does **context switching overhead** affect CPU efficiency?
3. Why can **SJF cause starvation**, and how can **aging** fix it?
4. Suggest how AI could improve scheduling in modern operating systems.